

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 7

7. THE HYDROGEN ATOM

• **z -projection of the angular momentum / Ex. 7.15**

What is the smallest value that ℓ may have if $\vec{L}$ is within $\theta = 10^\circ$ of the z axis?

Solution

Since θ denotes the angle that the angular momentum vector $\vec{L}$ makes with the z axis, then

$$\cos(\theta) = \frac{L_z}{L} = \frac{m_\ell}{\sqrt{\ell(\ell+1)}}.$$

Since L is quantized, there must be a minimal value necessary for its projection to be within a certain angular distance from the z axis. In the extreme case $m_\ell = \ell$ one finds

$$\cos^2(10^\circ) \leq \left[\frac{\ell^2}{\ell(\ell+1)} \right]$$

solving for ℓ yields

$$\ell = \left[\frac{1}{\cos^2(10^\circ)} - 1 \right] \approx 32.16$$

which one needs to approximate to $\ell = 33$ to find a solution closer that $\theta = 10^\circ$ to the z axis.

Note that while no pair of angular momentum components L_x , L_y and L_z can simultaneously be determined, this angle can be, because it can be expressed with only L_z and L^2 which commute with each other.

• **Zeeman effect in atomic spectra / Exs. 7.21 & 7.24**

Show that the wavelength difference between adjacent transitions in the normal Zeeman effect is given approximately by

$$\Delta\lambda = \frac{\lambda^2 \mu_B B}{hc}.$$

With this correction, the red line of the Balmer series for the Hydrogen atom ($\lambda = 656.5 \text{ nm}$) is observed to split into three spectral lines with $\Delta\lambda = 0.040 \text{ nm}$ between two adjacent lines when placed in a magnetic field B . What is the value of B if $\Delta\lambda$ is due to the energy splitting between two adjacent m states?

Solution

The change in the energy associated to the change of the spectral lines is obtained considering that $E = E(\lambda)$ as

$$E = pc = \frac{h}{\lambda} c$$

so that

$$dE = -\frac{hc}{\lambda^2} d\lambda, \tag{1}$$

and the energy splitting due to the magnetic field is given by

$$\mu_B B = |\Delta E| = \frac{hc}{\lambda^2} |\Delta\lambda|$$

meaning that the line splitting is given by

$$\Delta\lambda = \frac{\lambda^2 \mu_B B}{hc}.$$

Hence, the magnetic field that produces a splitting $\Delta\lambda = 0.040 \text{ nm}$ between two adjacent lines is given by

$$B = \Delta\lambda \frac{hc}{\lambda^2 \mu_B} = 1.99 \text{ T},$$

obtained using $hc = 1240 \text{ eV nm}$ and $\mu_B = 5.788 \times 10^{-5} \text{ eV T}^{-1}$.

• **Angular momentum commutation relations**

The angular momentum operators in quantum mechanics are defined analogously to classical mechanics by $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ using the position and momentum operators. The various angular momentum components obey the commutation relations

$$[L_\alpha, L_\beta] = i\hbar\epsilon^{\alpha\beta\gamma}L_\gamma,$$

where we used the commutator $[A, B] = AB - BA$, and $\epsilon^{\alpha\beta\gamma}$ is Levi-Civita's completely-antisymmetric tensor, defined so that $\epsilon^{\alpha\beta\gamma} = \pm 1$ for every *even/odd* permutation of $\{x, y, z\}$ and $\epsilon_{\alpha\beta\gamma} = 0$ if any index is repeated. Using the Heisenberg commutation relation for position and momentum $[r_\alpha, p_\beta] = i\hbar\delta_{\alpha\beta}$ (using the Kronecker delta: $\delta_{\alpha\beta} = 1$ if $\alpha = \beta$, $\delta_{\alpha\beta} = 0$ if $\alpha \neq \beta$), confirm the commutator relation

$$[L_z, L_x] = i\hbar L_y.$$

Analogously check that

$$[L_x, L_y] = i\hbar L_z.$$

and

$$[L_y, L_z] = i\hbar L_x.$$

Using the angular momentum commutation relations, check that $\mathbf{L}^2 = \sum_{\alpha=x,y,z} L_\alpha^2$ commutes with all the angular momentum components:

$$[\mathbf{L}^2, L_\alpha] = 0.$$

Solution

In order to derive these kinds of relations, a possible strategy is to substitute the expression of the angular momentum in terms of position and momentum operators as

$$[L_z, L_x] = [xp_y - yp_x, yp_z - zp_y] = [xp_y, yp_z - zp_y] - [yp_x, yp_z - zp_y] = [xp_y, yp_z] - [xp_y, zp_y] - [yp_x, yp_z] + [yp_x, zp_y].$$

Of all the above commutators, only two terms survive, so that

$$[L_z, L_x] = i\hbar(p_x z - xp_z) = L_y.$$

Strategy

In order to obtain the final result, one can exploit the following properties of commutators

$$[A, BC] = B[A, C] + [A, B]C \quad (1)$$

and

$$[AB, C] = A[B, C] + [A, C]B \quad (2)$$

Consider, e.g., the first commutator, $[xp_y, yp_z]$, and apply property (1) with

$$xp_y = A \quad \text{and} \quad yp_z = BC$$

so that

$$[xp_y, yp_z] = y[xp_y, p_z] + [xp_y, y]p_z,$$

and then apply property (2) with

$$[AB, C] = B[A, C] + [A, C]B$$

with

$$xp_y = AB \quad \text{and} \quad p_z = C \quad \text{or} \quad y = C$$

respectively for each of the terms. Finally (being careful not to interchange the order of the operators) the commutator is decomposed into the following

$$[xp_y, yp_z] = yx[p_y, p_z] + y[x, p_z]p_y + x[p_y, y]p_z + [x, y]p_y p_z$$

and according to the commutation relation for position and momentum operators

$$[r_\alpha, p_\beta] = i\hbar\delta_{\alpha\beta}, \quad \alpha, \beta \in \{x, y, z\}$$

the sole non-zero term, given by

$$x[p_y, y]p_z = -i\hbar xp_z.$$

The same procedure is applied to all other commutators, and the sole other non-zero term originates from the last one, yielding

$$[yp_x, zp_y] = i\hbar p_x z$$

thus

$$[L_z, L_x] = i\hbar(p_x z - xp_z) = i\hbar L_y.$$

All other commutation relations can be demonstrated analogously.

For the commutation relation with $\mathbf{L}^2$, we expand the commutator for $\alpha = x$:

$$[\mathbf{L}^2, L_x] = [L_x^2, L_x] + [L_y^2, L_x] + [L_z^2, L_x].$$

The first term vanishes, as every operator commutes with itself and its powers. Let us expand the second term using the commutation relation $[L_x, L_y] = i\hbar L_z$, implying $L_x L_y = i\hbar L_z + L_y L_x$:

$$[L_y^2, L_x] = L_y^2 L_x - L_x L_y^2 = L_y^2 L_x - (i\hbar L_z + L_y L_x) L_y = L_y^2 L_x - i\hbar L_z L_y - L_y (i\hbar L_z + L_y L_x) = -i\hbar (L_z L_y + L_y L_z).$$

Similarly expanding the third term we find

$$[L_z^2, L_x] = i\hbar (L_z L_y + L_y L_z),$$

hence these terms cancel, proving the commutation relation, the other components can be computed analogously.

Comment

Another way to derive the above commutation relations without relying on the properties of commutations of four operators is to directly evaluate the commutations in terms of the position and momentum ones. For instance

$$[xp_y, yp_z] = xp_y yp_z - yp_z xp_y.$$

The strategy is to take one of the two terms and commute the operators pair-wise in order to make the operators they have in common appear in the same order. In this specific case, for the first term

$$xp_y yp_z = x([p_y, y] + yp_y)p_z = x[p_y, y]p_z + xyp_y p_z = -i\hbar xp_z + xyp_y p_z.$$

For the second term, since the only non-zero commutator is $[r_\alpha, p_\beta]$ for $\alpha = \beta$, and can reorder the operators as follows

$$-yp_z xp_y = -xyp_z p_y = -xyp_y p_z.$$

Putting together two terms, after a cancellation, the result is

$$[xp_y, yp_z] = -i\hbar xp_z,$$

which is the same result obtained above.

Comment

Note that for the Hydrogen atom, the Hamiltonian simultaneously commutes with both L^2 and (conventionally) L_z operators, i.e.,

$$[H, L^2] = 0,$$

and

$$[H, L_z] = 0,$$

so that the Hydrogen eigenfunctions' angular part are simultaneously eigenstates of L^2 and L_z . However, since the components of the angular momentum operator along different axes do not commute, it is not possible to find a complete set of solution that are also simultaneously eigenstates of either L_x or L_y .

Comment: Heisenberg uncertainty of angular momentum

The general form of the Heisenberg uncertainty principle for two operators A and B states:

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

In general the right-hand side depends on the state, the reason it is a constant for the position-momentum relation, is that $[x, p] = i\hbar I$ is proportional to the identity operator I , so its expectation value is the same in any normalized state.

Using the commutation relations of angular momentum, we get the uncertainty relation

$$\Delta L_i \Delta L_j \geq \frac{\hbar}{2} |\langle L_k \rangle|,$$

which is true for $\{i, j, k\}$ being any permutation of $\{x, y, z\}$, so this is three relations.

Let us look at some specific cases (all of these are also true for any permutation of $\{x, y, z\}$):

- Assume the electron is in an eigenstate of L_z , meaning that $\Delta L_z = 0$. This implies that $\langle L_x \rangle = \langle L_y \rangle = 0$. This happens because the other angular momentum components are in a superposition of their eigenstates, such that the expectation value is zero.
- Assume the electron is in an eigenstate of L_z with $\langle L_z \rangle = \hbar m_\ell$. This implies $\Delta L_x \Delta L_y \geq \frac{\hbar^2}{2} |m_\ell|$, meaning that at least one of L_x and L_y has a high uncertainty, and neither of them can be exactly determined if L_z is highly polarized. Assuming that the orbital has total angular momentum $\ell > 0$, the maximum uncertainty happens in a state that is an equal superposition of $\pm\ell$ states, meaning $\Delta L_i \leq \hbar\ell$. Using this for ΔL_y , we find that $\Delta L_x \geq \frac{\hbar}{2} \frac{|m_\ell|}{\ell}$. This already shows that if L_z is in an eigenstate with $m_\ell \neq 0$, the other two components cannot be in eigenstates because they have a finite uncertainty.
- Assume the electron is simultaneously in an eigenstate of L_x and L_y , meaning that $\Delta L_x = \Delta L_y = 0$. The uncertainty relations imply $\langle L_x \rangle = \langle L_y \rangle = \langle L_z \rangle = 0$. Using the definition of the uncertainty, this also means $\langle L_x^2 \rangle = \langle L_y^2 \rangle = 0$. We know that

$$\langle \vec{L}^2 \rangle = \langle L_x^2 + L_y^2 + L_z^2 \rangle = \hbar^2 \ell(\ell + 1).$$

Using the previous results, we get $\langle L_z^2 \rangle = \hbar^2 \ell(\ell + 1)$. This is a contradiction because the largest eigenvalue L_z can have is $\pm\hbar\ell$, hence $\langle L_z^2 \rangle \leq \hbar^2 \ell^2$, showing that the electron cannot be simultaneously an eigenstate of two angular momentum components.